No Free Credit: How Much Simulation Evidence Can Replace Field Events in Zero-Failure Acceptance Testing of a Perception System

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Abstract

The credibility of simulation evidence is graded qualitatively or by uncertainty analysis, not in field events replaced. For a camera perception system under a zero-failure acceptance test, a fixed count of real events (59 for 95% confidence that at most 5% are missed) must pass without a miss. We prove that, without a sim-to-real link, independent simulation cannot shorten the test; that a larger error probability buys an exact number of events; and that a synthetic twin paired with each real event bounds the real miss rate by the twin miss rate plus discordance, which pays off only through a transfer bound priced in calibration scenes. On 48 fixed-camera scenes, real misses are rare (0.9%); at the pre-specified operating point the twin as built replaces no field event for the median camera (3 of 48 cameras save any), and a twin needing no real events of a new camera, if equally discordant (not built), would save at most 14 events each (20% target). The transfer bound stands at scene level but is withdrawn at site level (17 sites). Out-of-domain pass-through has two named modes: night and camera-jitter cameras are falsely accepted (4 of 6; 1 of 26, a real-shake scene).

Keywords: simulation credibility, validation, synthetic twin, zero-failure acceptance test, perception system, transfer bound

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1. Introduction

Simulation models are used more and more as evidence that a system may be accepted; regulators, for instance, have started to accept evidence obtained in silico [1]. When such evidence is credible has its own literature. Validation compares a model with real data for its intended use [2, 3, 4]; the ASME V&V 40 standard assesses the credibility a model needs through a risk-based framework [5, 6]; verification, validation and uncertainty quantification frameworks and Bayesian calibration attach uncertainty to model predictions [7, 8]; and the credibility of digital twins has been evaluated with evidence theory [9]. These frameworks express credibility qualitatively, as levels or scores, or through uncertainty analysis. None of them states how many field observations a credible model may replace in a given acceptance test.

A perception system on a fixed camera is a favourable case for this question: the context of use is the acceptance of a new fixed-camera installation within a declared domain of static scenes, and a composite model can keep the real background, illumination and detector while synthesising only the passing object, so operational validation of the composite model, for this decision, reduces to comparing the system's output on paired real and synthetic events. We study the question for one common test. A zero-failure (success-run) test accepts a unit when it passes a planned number of demands (independent trials; here, object passages) without failure [10, 11]. If each demand fails independently with probability p, a unit with $p > \varepsilon$ passes n demands with probability at most $(1-\varepsilon)^n$, so passing $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1-\varepsilon) \rceil$ demands without failure demonstrates $p \leq \varepsilon$ at confidence $1-\delta$: 59 demands at $\varepsilon = \delta = 5\%$. In reliability engineering this test is called a reliability demonstration test; we call it a *zero-failure acceptance test*. Its sample-size burden motivates test plans that borrow auxiliary information [12, 13].

The unit under test is a learned perception component: a fixed object detector on a fixed camera that samples frames periodically. A demand is the passage of an object (an *event*); a failure is an event on which no sampled frame yields a detection. Demands are rare and every new installation is a new unit, so 59 failure-free events must be collected per camera; for road vehicles the field exposure needed to show rare-failure reliability is prohibitive [14]. Simulation is the obvious remedy: digital twins already serve as platforms for testing path-planning algorithms [15], as virtual scenes for camera detectors [16], and as generators of synthetic sensor data with

a validation step [17]; virtual scenario testing is a leading proposed alternative to mileage-based testing of automated vehicles [18]. For perception, however, the simulation-to-reality gap is measurable and depends on conditions [19, 20], and performance in simulation need not predict performance in the field [21]. The credibility question is therefore quantitative: *how many of the n_0 field events can simulation evidence replace, and what form of guarantee survives when the evidence is transferred to an untested unit?*

This paper answers the question for the zero-failure acceptance test of a perception component. We use the *composite simulation model*, or *synthetic twin*, of an event: the real background of the event with the object replaced by a rendered one drawn on the ground-truth trajectory (Section 5).

1. *No free credit.* Without an assumption that links the simulator to reality, no valid procedure can use independent simulated data to shorten a zero-failure test, for any simulator and any amount of simulation. This makes precise, for success-run tests, a principle known from clinical borrowing [22], and relates to the limits on combining evidence in software dependability [23].
2. *The exact price of relaxing confidence.* Accepting error probability $\delta' > \delta$ when the simulator is wrong buys at most, and exactly, $n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta')$ demands: 8, 14, 22 and 27 of the 59 for $\delta' = 7.5$, 10, 15 and 20%.
3. *A paired-twin bound and a fleet-level transfer bound.* A synthetic twin of each real event bounds the real failure probability by the twin failure probability plus a discordance rate, with no assumption; on one camera this costs more real events than the plain test (72–122). Calibrated once across scenes, the discordance transfers to an untested camera as a population-average bound whose price is the number of calibration scenes, not the number of events.
4. *An exchange rate on public benchmarks, and a named failure mode.* On 48 fixed-camera scenes the twin as built saves no real event for the median camera (3 of 48 cameras save any); a twin run to convergence would save at most 14 real events per camera, if the discordance calibrated with paired twins also held for the twins of new cameras (not verified), at the operating point fixed in advance, where real failures are rare (0.88%) and the certificate is therefore untested. The transfer certificate stands at scene level but is withdrawn when the unit is the site (17 sites). When out-of-domain cameras are judged alone, they

pass through: 4 of 6 night cameras and, at a fixed sampling phase, 1 of 26 camera-jitter cameras (a scene with real shake) are falsely accepted; we call this *out-of-domain pass-through*, with two named modes, night and camera jitter.

Simulation credibility is today scored qualitatively or by uncertainty analysis; to our knowledge, no result states how much field evidence a credible twin may replace in a zero-failure acceptance test, nor how that credit transfers to an untested unit. This paper gives a finite-sample answer (none without an assumption; an exact price for relaxing confidence; a fleet-level bound whose price is scene count) and measures it on 48 fixed-camera scenes.

2. Related work and research gap

Table 1 positions the paper against four bodies of work.

(a) Simulation validation and credibility. Verification and validation of simulation models is an established discipline that separates conceptual model validity, verification, operational validity and data validity [2, 3]. In validation the relation between computation and experimental data is the issue, and predictive capability goes beyond it [4, 7]; Bayesian calibration learns the unknown parameters of a computer model from physical observations of the system [8]. ASME V&V 40 provides a risk-based framework for assessing the credibility of a computational model through predefined credibility activities [5, 6], and a recurring difficulty is that validation cannot be performed in a setting that closely resembles how the model will be used [24]. Regulatory use of in silico evidence has made such credibility assessment a practical matter [1]. For digital twins, credibility has been evaluated with evidence theory [9]; twins are verified against physical experiments [36, 37], driving-simulator validation studies tie the credibility of results to scenario fidelity [38], and a review of manufacturing twins found little work that performed both verification and validation [39]. *What is missing:* credibility is expressed as a level, a score or a predictive uncertainty; none of these is converted into the number of field observations a credible model may replace in a given acceptance test, and none bounds how that replacement carries over to a unit whose own validation data are absent.

(b) Simulation-assisted testing of automated and perception systems. Digital twins serve as platforms for testing and optimising path-planning algorithms [15], as virtual tunnels for a camera flame detector [16] and as generators of synthetic spectra with a validation step [17]; a multifidelity simulation

audit screens the dynamic executability of continuum-robot motions and reports accepted cases out of randomized trials [40]; virtual scenario testing has been proposed in place of mileage-based testing of automated vehicles [18], and model validation for automated driving names extrapolation to untested conditions as a major challenge [25]. The simulation-to-reality gap of camera object detectors has been measured across day, night, fog and rain [19] and for drone detection [20]; synthetic images used to test driving perception have a distribution mismatch that can bias test results [41], perception digital twins reproduce field data only approximately [42], statistical model checking evaluates perception in simulation only [43], simulation performance can fail to predict field performance [21], and a statistical foundation for aligning synthetic and real testing outcomes has been called for [44]. Closest to our question, Sim2Val uses simulator outputs as control variates to reduce the number of real samples needed for a confidence bound on mean performance (CoRL 2025, arXiv preprint) [26], SCAPE combines limited real rollouts with simulation proxies for policy evaluation (preprint) [45], multi-fidelity fusion weights simulation by its fidelity gap in reliability prediction [46], and prediction-powered inference combines many proxy outcomes with few real labels through a paired rectifier [27, 47]. *What is missing:* these methods estimate a mean performance or reduce its variance; none treats the zero-failure regime, where the quantity to bound is a rare failure probability and the rectifier itself would cost almost n_0 real events per unit.

(c) Zero-failure test planning. The zero-numerator bound and the rule of three are classical [10, 11], and the Bayesian zero-failure (BAZE) test sets the required test time through a prior on the failure rate [28]. Later work designs sample sizes by assurance [29], studies the limits of Bayesian zero-failure plans [48] and plans success–failure verification and qualification tests with subsystem, Monte-Carlo or expert information [12, 13, 49]. For software and autonomous systems, conservative Bayesian inference uses prior knowledge, including evidence from previous similar systems, without optimistic bias [30, 50, 51], and reusing the reliability of an older system as a prior is optimistic rather than conservative [52]. Bayesian borrowing quantifies the worth of a prior through its effective sample size [53] and discounts historical data through power or commensurate priors [54, 55]; conservative frequentist bounds exist when the test profile differs from the operational profile [56]. *What is missing:* a frequentist statement of how much the zero-failure sample size can be reduced by simulation rather than prior-system evidence, and at what price.

(d) Finite-sample guarantees and assurance for learned perception. Conformal prediction gives distribution-free finite-sample guarantees [57], calibrated risk control extends them to general losses [31], and they have been applied to object detection [32], keypoint-based pose estimation [58], perception for safe navigation [33] and anomaly detection in cyber-physical systems [34]. Reliability of machine-learning components is also modelled directly, through multi-version architectures [59], safety-assurance cases for object detectors [35] and failure-free-operation definitions under an operational profile [60]. *What is missing:* these guarantees are calibrated on real data from the deployment distribution; they do not say how much of that calibration data may be synthetic, nor how a guarantee transfers to a unit whose own data are absent.

3. Acceptance-test setting

Unit, demand and failure. The unit under test is a fixed camera s with a fixed detector. The camera observes *events*: maximal runs of frames that contain the target object. It samples frames with period K and an unknown phase φ, and runs the detector on the sampled frames. An event is *missed* ($M = 1$) if no sampled frame is a hit; with H the hit frames of an event and φ uniform, $\mathbb{P}(M = 1) = 1 - |H \bmod K|/K$. The miss probability of camera s is p_s, and the acceptance goal is the claim $p_s \leq \varepsilon$ at confidence $1 - \delta$. $U(x, n)$ denotes the one-sided $(1 - \delta)$ Clopper–Pearson upper bound for x failures in n demands. We call an accepted assurance claim of this form a *certificate*. In the language of acceptance sampling, δ is the consumer's risk; the producer's risk is reported as the power of a plan at the observed miss rate. All claims concern the miss probability averaged over the uniform sampling phase; a camera deployed with one fixed, unknown phase is covered only on average over phases.

Simulated evidence and twins. A simulator produces data $Y \sim Q$. A *paired twin* renders a synthetic counterpart of each real event and runs the same detector at the same sampling phase, giving M'. We write $q = \mathbb{P}(M' = 1)$ and $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch), the *discordance*. Every rate in this paper is per event; with three sprite variants per event, the twin catches an event if at least two variants do, and an event counts as discordant if it is discordant at one or more sampling phases (worst phase; an integer count, conservative with respect to the phase-averaged rate, which we give in parentheses). Table 2 collects the notation.

Random model. Scenes are drawn from a population; given a scene, events are drawn i.i.d. from the scene's event law and each receives an independent uniform phase; given the event and the phase, the real and twin outcomes are determined (fixed detector, fixed sprite seeds). Every per-event indicator below—real miss, twin miss, discordance at a random phase, and the worst-phase indicator—is therefore a Bernoulli variable, independent across events given the scene. In the audit, the "truth" of a scene is its empirical event law, resampled with replacement.

4. What simulated evidence can buy: four results

The four results below are stated in words; Appendix A gives the formal statements, the assumptions and the proofs, each with a numerical check.

Result 1 (no free credit). Suppose the simulated data are independent of the real demands and no assumption links the simulator to reality, so that for every simulator law Q both a perfect unit ($p = 0$) and a unit just worse than the target ($p \downarrow \varepsilon$) are possible. Then any acceptance procedure that is valid at confidence $1 - \delta$ accepts a perfect unit with probability at most $\delta/(1-\varepsilon)^n$ when it uses n real demands, whatever the simulator and however much simulation it uses. Accepting a perfect unit with certainty therefore still needs $n \geq n_0(\varepsilon, \delta)$ real demands. The reason is that the simulated data have the same law under both hypotheses, while an all-success real sample is possible under both. This makes precise, for the success-run test, the principle that external evidence cannot shorten a test whose validity must hold for every external law [22], and the related limit on combining evidence for ultra-high dependability [23]. A paired twin is built from the real events and is therefore *not* an independent simulator; pairing is what Result 3 exploits.

Result 2 (the exact price of relaxing confidence). If the test may accept a worse unit with probability $\delta' > \delta$ when the simulator is wrong—that is, if one accepts a higher consumer's risk—the real demands can be reduced by at most

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta'),$$

and this reduction is attained by a two-tier plan: run the zero-failure test with $n_0(\varepsilon, \delta')$ demands when a simulator test passes and with $n_0(\varepsilon, \delta)$ otherwise. The plan keeps confidence $1 - \delta$ only if a simulator of a worse-than-target

unit passes with probability at most π^*: 6.1%, 3.0%, 1.5% and 1.0% for $\delta' = 7.5$, 10, 15 and 20% at $\varepsilon = \delta = 5\%$. This requirement is itself a link between simulator and reality, so Result 2 is best read as the price of lowering the confidence, not as a free credit. The frequently used continuous form $\lfloor \ln(\delta'/\delta)/(-\ln(1-\varepsilon)) \rfloor$ is only a lower bound on c. Table 3 gives the numbers.

Monte-Carlo runs match the closed form within 0.29 percentage points on a 40-point grid. Used as an independent simulator (its events resampled independently of the real draws), the twin event pool passes the simulator test in 73.3% of runs, so the credit is only partly realised (73%), and the false-acceptance probability at $p = \varepsilon$ is δ' up to the pass probability: the credit is bought exactly with the extra error. An off-the-shelf synthetic benchmark (the synthetic part of the BMC benchmark [61]; 30 events long enough for the operating point) never passes at $\varepsilon = 5\%$ (0%): it has too few synthetic events to bound its own miss probability below 5%. At $\varepsilon = 20\%$ it passes in every run and buys the same credit as the twin (supplementary Table S11), which shows that passing the simulator test is no evidence of fidelity. Monte-Carlo standard errors are about 0.1%.

Result 3 (paired-twin bound). For any pairing of real and twin events, the real miss probability is at most the twin miss probability plus the discordance,

$$p \le q + D,$$

with no assumption on the simulator (the bound splits the real misses into those the twin also misses and those it catches). On a single camera the bound does not save demands: if $q \le q_U$ holds with confidence $1 - \delta/2$ and $D \le \varepsilon - q_U$ is demonstrated by a zero-discordance run at level $\delta/2$, the camera needs $\lceil \ln(\delta/2)/\ln(1-\varepsilon+q_U) \rceil$ paired events, i.e. 72, 91 and 122 for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$, all above $n_0 = 59$. A twin built with a pessimistic (shorter) event length remains valid: its miss probability can only grow and its discordance only shrink. On the fixed-camera twin, shortening every event by 50% raises the twin miss rate at OP* from 1.37% to 2.36%. With a uniform phase the discordance probability of an event is $|R' \setminus R|/K$, where R, R' are the residues modulo K of the real and twin hit frames; the worst-phase indicator $\mathbf{1}\{R' \setminus R \neq \emptyset\}$ dominates it, so the bounds of Result 4 also hold for the integer worst-phase counts we report.

Result 4 (transfer across scenes). Here *calibration* means collecting paired validation data to estimate the discordance; no parameter of the compos-

ite model is tuned. Let m calibration scenes contribute $N = \sum_s n_s$ paired events, X of them discordant, and let J scenes show at least one discordant event. (a) *Population bound (the transfer guarantee).* If the scenes are drawn independently from a scene population (events within a scene may be dependent), the population-average discordance of a new camera is at most $U(J, m)$ with confidence $1 - \delta$, so its expected miss probability is at most its expected twin miss probability plus $U(J, m)$. The bound counts scenes, not events; it needs no independence of events within a scene. Its price is the number of scenes: with $J = 0$, $U \leq 10\%$ needs 29 scenes and $U \leq 5\%$ needs 59. With the 48 fixed-camera scenes of this study, $J = 0, 1, 2, 3$ give $U(J, m) = 6.1\%$, 9.5%, 12.5% and 15.4% at level δ and 7.4%, 11.1%, 14.3% and 17.2% at level $\delta/2$: a single discordant scene costs about three percentage points, and a bound below $\varepsilon/2 = 10\%$ tolerates one discordant scene at level δ and none at $\delta/2$. Adding events to existing scenes does not tighten it; only new scenes do. (b) *Fleet bound (calibrated cameras, independent events).* For the calibrated cameras, the event-weighted discordance is at most $U(X, N)$ with confidence $1 - \delta$ (for $\delta \leq 1/4$), however heterogeneous the cameras are; hence their event-weighted miss rate is at most their twin miss rate plus $U(X, N)$. Applying it requires a separate bound on the twin miss rate, with its own share of δ; where real misses exist, direct Clopper–Pearson on them is at least as informative. The proof compares the Poisson-binomial count with a binomial of the same mean [62] and uses a lower bound on the probability that a binomial reaches its mean [63], applied to the complementary count. This bound assumes that events are independent given the scenes, so positive dependence within a scene makes it optimistic, and it says nothing about a new camera. (c) *Per-camera limit.* A per-camera statement, "$p \leq q + t$ except on a set of cameras of mass γ", follows if every scene has at least $n_{\min}$ events, with $\gamma = U(J, m)/(1 - (1 - t)^{n_{\min}})$. Table 4 turns this into a data requirement: for $\gamma = 5\%$ at least 59 scenes, each with tens to hundreds of events. No public benchmark we know offers this, so the per-camera form is reported as a limit.

Acceptance procedure. The procedure evaluated in Section 6 is: (1) declare the domain and pair every real event of m calibration scenes from it with its twin; count discordant scenes J; (2) *domain rule*: if $U_\delta(J, m) > \varepsilon/2$, withdraw the transfer claim for the whole domain (a design rule for go/no-go, not a theorem); (3) for a new camera of the declared domain, run its twin until its own miss probability is bounded, q_U, at level $\delta/2$ (a twin *run*

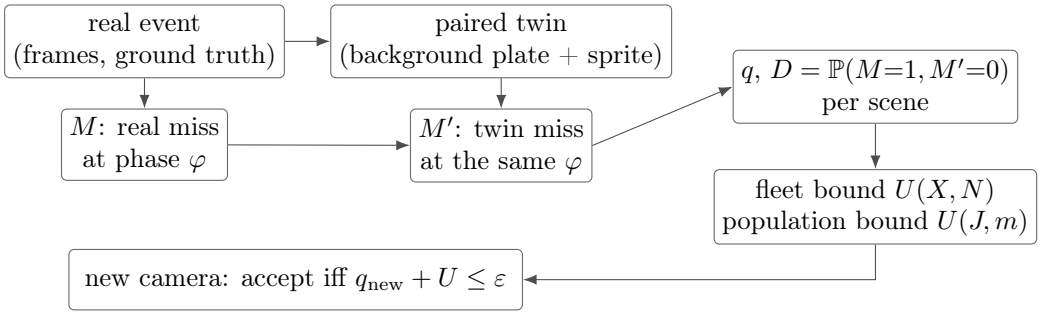


Figure 1: The acceptance test with a paired twin. Each real event is paired with a synthetic twin that replaces the object and keeps the scene; both are evaluated by the same detector at the same sampling phase. Discordances calibrated across scenes bound the miss probability of a new camera through $p \leq q + D$.

is one rendering of one synthetic event in one sprite variant; runs are treated as independent, which is optimistic when runs share a trajectory); (4) accept the camera with no real demand if $q_U + U_{\delta/2}(J, m) \leq \varepsilon$, otherwise fall back to the zero-failure test (Fig. 1). By Result 4(a) and a union bound, with confidence $1 - \delta$ the population-average discordance is at most $U_{\delta/2}(J, m)$. Among *accepted* cameras the average discordance can be larger, by at most the factor $1/\mathbb{P}(\text{accept})$, because acceptance selects cameras with a low twin miss rate; the average miss probability of accepted cameras is therefore at most $\varepsilon - U_{\delta/2} + U_{\delta/2}/\mathbb{P}(\text{accept})$. At OP*, where 95.8% of the static cameras (46 of 48, phase-averaged) are accepted, the inflation is 0.3 percentage points; out of domain it is unbounded (Section 6.3). The acceptance is *not* a per-camera guarantee. In the audit we nevertheless count, per camera, acceptances with $p_s > \varepsilon$ ("false acceptances") as a diagnostic of how the average guarantee behaves on individual cameras.

5. Experimental protocol

Data. The corpus is rebuilt from the raw detections and pixel ground truth of CDnet 2014 [64] and LASIESTA [65] (frames in true temporal order), with the synthetic part of the BMC benchmark [61] as an off-the-shelf simulator; the detector-defined corpus has 3,170 events from 101 real videos. For the twin we use ground-truth events: a frame is a *hit* if a detector box covers at least half of the largest ground-truth component or overlaps its bounding box with IoU ≥ 0.3, and an event counts if its largest object has at least 256 pixels

(primary definition; the original definition keeps every ground-truth event). Scenes are assigned to tiers by the category labels of the two datasets. The *static* (main) domain is every CDnet calibration scene outside the categories *PTZ*, *cameraJitter*, *nightVideos* and *turbulence*, and every LASIESTA calibration scene outside the categories SM (simulated camera motion) and MC (moving camera): 48 scenes (28 CDnet, 20 LASIESTA; 161 static-domain events of any duration, 114 of at least 32 frames). The other tiers are night (6 scenes), turbulence (4 scenes; only 2 were run and they hold 2 events, so the tier is reported but not certified) and camera jitter (26 scenes: CDnet *cameraJitter* and LASIESTA SM); the 4 pan-tilt-zoom and 4 moving-camera scenes are excluded, because the twin assumes a static background. This definition was fixed after the initial version of this study, whose static domain was wider; no twin or detector was re-run, and the supplement compares the static domain of the three versions (Table S16).

Detector and twin. The detector is a fixed single-stage segmentation network (YOLO family [66]; yolo26s-seg [67], confidence 0.25, input 640 px, CPU). Per scene, a background plate is the per-pixel median of up to 60 frames without ground-truth foreground. For each real event and each of three sprite variants (fixed seeds), the twin removes the real object (mask dilated by 3 px), fills it from the plate, and draws a synthetic object on the ground-truth trajectory: a vehicle cut from Virtual KITTI 2 [68], chosen by aspect ratio, or a procedurally dressed human rendered with MPFB [69] with a four-frame walking cycle mirrored to the motion direction. The sprite height equals the ground-truth box height, its width is clamped to $\pm 10\%$ of the box width, its brightness follows the frame luminance, and a soft shadow is added. Only the object is synthetic; scene, illumination and background clutter are real. The detector runs on twin frames until the twin hits cover all residues modulo 48, which is exact for every K dividing 48. The median detector confidence on twin hits is 0.78 (real: 0.87). Building the twins of the 48 static scenes took 14.1 CPU-hours (median 0.20 per scene) on a laptop CPU.

What the twin consumes. The paired twin uses, per camera, empty frames (for the plate) and the annotated trajectories of the camera's real events (for sprite placement). Calibration therefore costs annotated events: 114 at OP*, i.e. 2.4 per static camera when amortized over the domain. For a *new* camera, the exchange rates below assume that the twin miss rate q_{new} can be measured without that camera's real events, e.g. with synthetic trajectories drawn on

its empty background. We did not build such trajectory-free twins; we use the paired q_s in their place. The discordance of a trajectory-free twin has not been calibrated, so the reported savings are conditional on the paired-twin discordance carrying over, and are upper bounds.

Operating points. An operating point is $(\varepsilon, d_{\min}, K)$ with events of at least $d_{\min}$ frames. The primary point $\mathrm{OP}^* = (20\%, 32, 16)$ at $\delta = 5\%$ was fixed by a rule chosen before the twin was built (the largest K for which direct certification needs at most 150 events at 80% power and the 44 calibration scenes available at that time give a population bound of at most $\varepsilon/2$), using the pooled real miss rate of all CDnet events of every category (11.3% over 161 events of all CDnet categories, a population distinct from the static events above, 112 events for direct certification). In the static domain the real miss rate at OP^* is only 0.88%; we report this, keep OP^*, and add the sensitivity points OP-A $(20\%, 32, 8)$ and OP-C $(10\%, 32, 1)$ and the full grid $d_{\min} \in \{1, 4, 8, 12, 16, 24, 32\}$, $K \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 48\}$. The night and turbulence tuning criteria and their held-out scenes were fixed before tuning. These choices are dated in internal project records but were not publicly registered. Six choices were made after results were available and are disclosed: (i) the primary event definition (largest object at least 256 pixels) was adopted after the initial twin run, with the original definition kept as a co-reported sensitivity (Table 11); (ii) the list of LASIESTA calibration scenes, before the split into tiers, was rebuilt (from 20 to 48 sequences, of which 20 fall in the static domain) after correcting the frame order of that dataset, before the twin was built; (iii) the two held-out turbulence scenes were not run; (iv) the CDnet *cameraJitter* scenes and (v) the LASIESTA SM and MC scenes were removed from the static domain after static-domain results were available; and (vi) a second operating point with more real misses was examined after inspecting the data (Section 6.3). The operating point OP^*, the audit design and the night criterion were fixed before the static-domain twin results; everything else in Section 6 is exploratory.

Why a paired twin. A reweighting alternative [70] needs overlap between synthetic and real events. A classifier separating BMC events from real events reaches a median AUC of 0.64 with temporal features, 0.94 with geometry and 0.77 with detector confidence alone (all targets); the gap lies mostly in geometry, so a reweighted synthetic benchmark cannot stand in for the real event distribution, and the twin is matched on geometry by construction.

Audit. Leave-one-scene-out (LOSO): each scene is certified with the bound computed from the other scenes. Monte Carlo (10,000 draws): calibration scenes are resampled from the static domain and a new camera is drawn from it; a false certificate is a certified camera with $p_s > \varepsilon$ (the resampled calibration set may contain the test camera, which makes this check optimistic; the LOSO audit does not have this defect); the "truth" of a camera is its own event pool, exact over phases. Two must-fail scenarios are run (Section 6.3). All randomness uses seed 42.

6. Results

6.1. Certificates by tier

In the static domain, 0 of 114 events at OP* are discordant; hence $U(X, N) = 2.59\%$ and, with $J = 0$ of 48 scenes, $U(J, m) = 6.05\%$ (Table 5, Fig. 2). Both are below $\varepsilon/2 = 10\%$. In the camera-jitter tier, 3 of 62 events are discordant, all in the CDnet scene *traffic*, which has real camera shake; the LASIESTA scenes with simulated camera motion have no discordant event. At night, 12 of 25 events are discordant (48%; phase-averaged 8.21 as a mean over variants, 8.31 under the majority rule) and 5 of 6 scenes; pooling all tiers raises the bounds to 11.7% and 15.4%.

6.2. Exchange rate

Fig. 3 and Table 7 show the real events a static camera saves when the twin replaces direct certification, across 70 operating points. A camera is certified with no real event when its twin miss rate plus the transfer bound (both at $\delta/2$, the bound computed without the camera) stays below ε. At OP* the direct test needs $n_0 = 14$ events and, because the static-domain miss rate is low, no larger plan; a twin run to convergence saves those 14 events at 46 of 48 cameras (phase-averaged; 44 at any fixed phase), at a measured cost of 0.069 CPU-hours per saved event (conditional on the discordance carrying over, Section 5; no accepted camera has $p_s > \varepsilon$ at OP*, but acceptances were not audited at every grid cell). **The twin as built saves no real event for the median camera (3 of 48 cameras save any); the saving requires about 28 twin runs per camera.** With three runs per event, as built, the twin's own miss rate cannot be bounded tightly enough, and at most 4 cameras save anything at any grid point. The saving grows with the real miss rate, because direct certification becomes expensive: it reaches hundreds of events per camera, and over a thousand at the most extreme cells of the

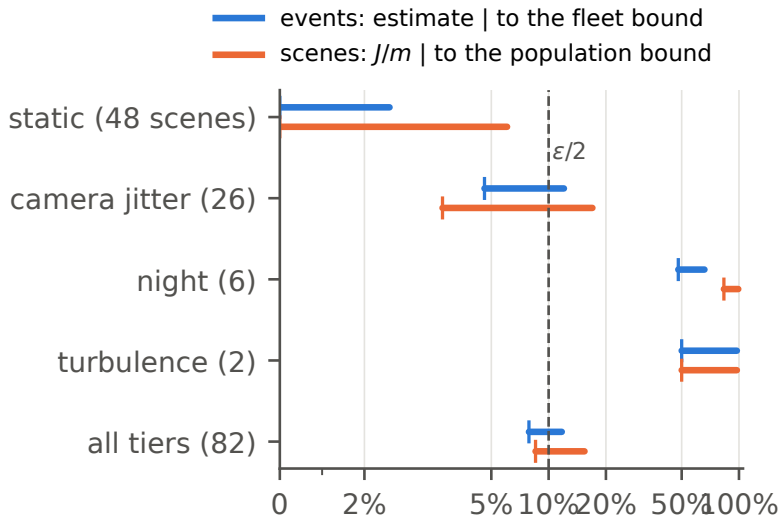


Figure 2: Certificates by tier at OP* (event unit, worst phase). Blue: discordant events per event and the fleet bound; orange: discordant scenes per scene and the population bound. Dashed: $\varepsilon/2$.

supplementary grid, with $\hat{p}$ above 10%, while at $K = 48$ and closer to ε the twin route saves nothing (Table 7). The target $\varepsilon = 20\%$ is a demonstrator, not a safety target. At $\varepsilon = 5\%$ the domain rule requires a population bound of at most $\varepsilon/2 = 2.5\%$, which needs at least 119 calibration scenes even with no discordant scene; with 48 scenes the twin route is not available at $\varepsilon = 5\%$, whatever the twin budget. For the calibrated cameras themselves, direct Clopper–Pearson on their pooled real misses gives 4.1%, close to the fleet discordance bound 2.59%: the fleet certificate adds little where real data exist, and its role is as the building block of the transfer.

6.3. Audit and must-fail scenarios

Is the certificate right because it is empty? Table 8 cross-tabulates real and twin outcomes. In the static domain at OP* almost every event is caught by both: the events with a real miss at some phase number 1 of 114, in 1 of 48 scenes, and the twin also misses 100% of the expected real misses, i.e. one event. The zero-discordance result is therefore mostly a statement about a detector that rarely misses in this domain, not a test of the twin. At night the twin sees almost none of the real misses.

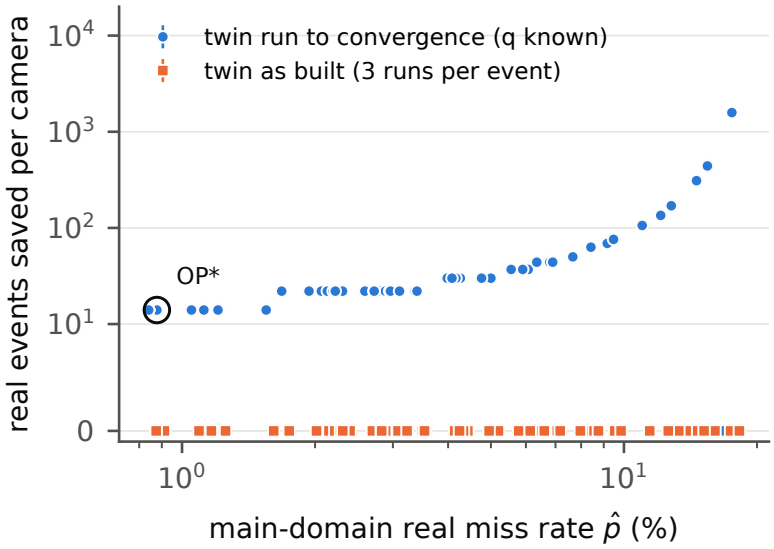


Figure 3: Exchange rate: real events saved per static camera (median and interquartile range over 48 cameras) against the static-domain real miss rate, one point per $(d_{\min}, K)$ at $\varepsilon = 20\%$. Blue: twin run to convergence; orange: twin as built (three runs per event). Vertical axis: symmetric log (linear below 10). Circled: OP*.

LOSO and Monte Carlo. At OP*, the number of static scenes whose discordance exceeds the bound computed without them is 0 of 48 (Fig. 5); the Monte-Carlo false-certification rate is 0.0%. Because real misses are rare at OP*, this audit cannot reveal much: only 1 static scene has $p_s > \varepsilon$ there (the one holding the single missed event), and it is refused, so the 0% rate rests on that one scene. A scene above its leave-one-out bound would still be compatible with a population-average guarantee. The informative tests are must-fail 2 and the camera-jitter check below, which can fail and do.

Must-fail 1: temporal. At $K = 48$, 3 static scenes have $p_s > \varepsilon$ and the procedure refuses them in 100% of runs; but most of these misses are temporal (events shorter than K, which the twin reproduces by construction); the number of scenes that exceed ε through detector misses is 1 (Table 9, top).

Must-fail 2: detector. We add k of the 6 night scenes to the calibration as if they were static-domain scenes; at OP* every real miss of these scenes is a detector miss. The transfer bound rises with k (Fig. 4, Table 9, bottom): from 6.1% to 18.5% at OP*, crossing $\varepsilon/2$ for some subsets at $k = 2$ and for

all at $k = 3$, so the domain-level certificate is withdrawn; no static camera is ever falsely certified (0). *Out-of-domain pass-through:* the per-camera rule does not protect the night cameras themselves. The night twin misses few events (twin miss rate 4.8% averaged over phases, 16.0% at the worst phase; its appearance model only dims the sprite), so night cameras judged with a bound computed from the static scenes are accepted: all 6 under the phase-averaged rule, 5 of 6 at the worst phase; 4 of 6 night cameras have $p_s > \varepsilon$ and are falsely certified at OP* (Table 9, middle; the same count holds at the worst phase, with a different set of four cameras, supplement). Each night miss rate rests on 2 to 10 events and is uncertain; the finding is that the acceptance rule admitted these cameras although their observed miss rates exceed ε and their twins miss few events, not a precise estimate of their miss rates.

Camera jitter. The same pass-through occurs for a shaking camera, but it rests on a single scene. Judged by the twin route calibrated on the static scenes (worst-phase convention, Section 6.4), 26 of the 26 camera-jitter cameras are accepted and 1 of them is falsely accepted: *traffic*, a CDnet scene with real camera shake, whose worst-phase real miss rate is 37.5% (per-camera table in the supplement). Its twin catches every event at every phase, because the synthetic object is drawn on a stabilised plate while the real frames shake; under the phase-averaged reading this camera stays below ε (Table 9, camera-jitter rows), so this failure appears only for a fixed deployment phase. The LASIESTA scenes with simulated camera motion produce no discordance and no false acceptance, so they add little evidence either way about real shake. Under the same worst-phase rule 4 of 6 night cameras are falsely accepted at OP*.

A post-hoc operating point. To see the certificate under more real misses, the point $(d_{\min}, K) = (16, 16)$ was examined after inspecting the data; it fails the criterion we fixed before recomputing it on the static domain (both bounds at most $\varepsilon/2$ and no false acceptance), because its population bound is 12.5%, and it is not reported further.

6.4. Fixed-phase and site-level checks

Worst phase. A deployed camera keeps one sampling phase. Counting an event as missed if it is missed at any phase, for the real event, the twin (majority of variants) and the discordance alike, keeps the bound valid for every fixed phase, because an event missed at some phase is either missed by the twin or discordant at that phase. At OP* the static-domain real and

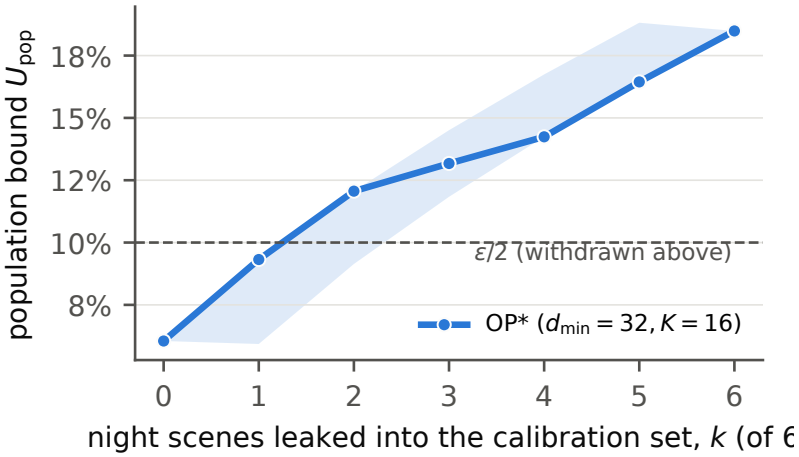


Figure 4: Must-fail 2: the population bound when k night scenes leak into the static calibration set (median over the $\binom{6}{k}$ subsets, band: min–max). The domain certificate is withdrawn above $\varepsilon/2$. Individual night cameras are still certified when judged alone (out-of-domain pass-through, text).

twin miss rates become 0.9% and 3.5% (phase-averaged 0.9% and 1.4%), while the discordance counts and both bounds are unchanged (Table 5). The direct plan then needs 14 events, and a twin run to convergence accepts 44 of 48 cameras, saving 14 events each (Table 6); the number of accepted static cameras with a worst-phase real miss rate above ε is 0.

Sites. Scenes filmed at the same site share background, viewpoint and illumination, so the scene may be too fine a unit of independence. Grouping the static scenes into 17 sites (CDnet challenge categories and LASIESTA sequence groups, a coarse proxy for sites), the number of sites with a discordant event at OP* is 0, yet with so few units the population bound is $U(J_{\text{site}}, m_{\text{site}}) = 16.2\%$, above $\varepsilon/2$: under the domain rule the transfer claim is withdrawn at site level. The leave-one-site-out audit finds 0 sites above their bound at OP*. The transfer certificate therefore stands at scene level and is withdrawn at site level; it holds only if scenes, not sites, are the independent units.

6.5. Failure modes

At night the twin sees few real misses: it predicts 8% of the expected missed events, and the tier bound is 99.1%. Fig. 6 shows why: the appearance

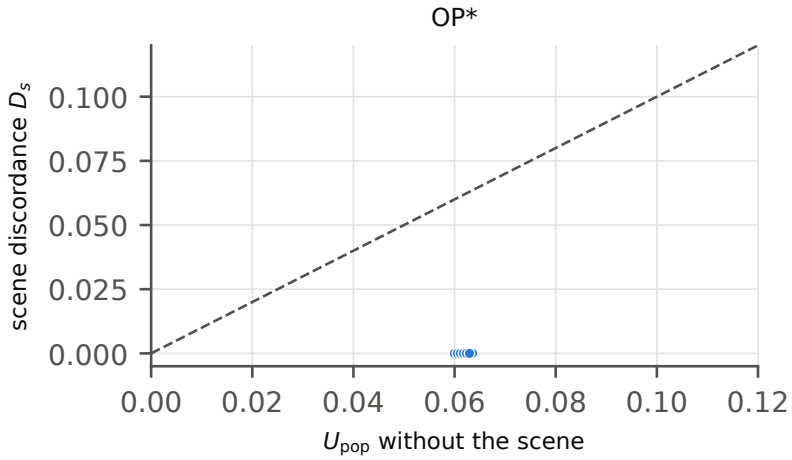


Figure 5: Leave-one-scene-out: per-scene discordance against the population bound computed without the scene. Points above the diagonal exceed the bound; the proposition bounds the population mean, not each scene.

model scales sprite brightness with the scene, but has no headlights, glare or sensor noise. An attempt to fix this was specified in advance: night-specific appearance profiles were tuned on four night scenes and judged on two held-out scenes, with success defined as a median held-out per-scene discordance of at most 5%. The best profile barely moved the tuning discordance (47.2% to 47.0%, phase-averaged) and the held-out median was 23.2% on 14 events, so the criterion failed, tuning was stopped, and night is declared outside the domain. Camera jitter is the second named failure mode, observed in one scene with real shake: the twin is drawn on a stabilised background plate, so it cannot reproduce the misses caused by camera shake (over the tier, twin worst-phase miss rate 0.0% against a real 4.8% at OP*; Table 5). One scene does not show how often real shake produces this failure. The turbulence tier holds 2 events at OP* and is not certified.

6.6. Sensitivity

Event definition. Keeping every ground-truth event (222 instead of 161 static events of any duration) leaves OP* almost unchanged: 116 events, 0 discordant, $U(X, N) = 2.55\%$, $J = 0$ (Table 11). *Operating point.* At OP-A and OP-C the static-domain miss rate is 0.88% and 0.88%, and no static-domain event is discordant. *Twin details.* The majority rule over sprite



Figure 6: Synthetic person sprites (CC0) on a neutral background under the brightness gain used for day and for night scenes. Dimming alone does not reproduce night imaging, which explains the low twin miss rate at night.

variants (1, 2 or 3 of 3) does not change any static-domain count. Table 10 compares twin and real hit rates on the same frames: the twin is pessimistic at every aspect error except one vehicle bin, and both hit rates fall when the ground-truth box has an unusual aspect (occlusion, objects entering the frame). Clamping the sprite width, which affects 70% of twin frames (more than the frames with an aspect error above 10% in Table 10, because 1-pixel rounding also triggers the clamp), lowers the twin hit rate by only 3.0 points more than the real hit rate on the same frames (631 event–variant pairs): it inflates q, not D. The median detector confidence on hits is lower for the twin than for real objects, especially for persons. Full tables are in the supplement.

7. Discussion and limitations

A narrow window. Simulation pays in events only when direct certification is expensive, that is, when the real miss rate is not far below ε; there the twin must also be accurate, and discordances make the transfer bound grow. When the miss rate is low, n_0 is small and there is little to save. The grid (Table 7) shows both ends: 14 events at OP*, hundreds at some operating points with $\hat{p}$ above 10%, and nothing where $\hat{p}$ comes close to ε.

Time. The waiting time saved at camera s is $n_{\text{saved}}/\lambda_s$, where λ_s is the event rate in deployment; we do not estimate it, because the benchmarks publish no frame rates and their clips are dense in events. If one CPU-hour

is valued like one hour of waiting, the twin breaks even when $\lambda_s < 1/c \approx$ 14 events per hour at OP*, with c the CPU-hours per saved event; this equivalence is an assumption, not a measurement.

Domain. The transfer bound is a population-average guarantee: it withdraws the domain certificate when out-of-domain scenes enter the calibration, but it does not detect an out-of-domain camera. The domain must therefore be declared and checked by other means (for example, illumination statistics), and a per-camera guarantee requires the data of Result 4(c). Three items are left for future work: a twin that needs no annotated real trajectory of the new camera, with its own calibrated discordance; a pre-specified in-domain stress test with cameras whose real miss rate exceeds ε; a verification of the compositing pipeline by a null twin that reinserts the real object and must reproduce the real outcomes; and a domain check (for example a per-camera discordance sentinel on a few real events) to stop out-of-domain pass-through.

Data and twin. No static scene has 59 or more qualifying events, so per-camera ground truth is the scene's own finite event pool; the twin builds its plate from real empty frames and therefore needs a static background; the sprite library lacks elevated viewpoints for vehicles; and we did not use a full 3-D simulator. The operating point was fixed from the pooled miss rate of all CDnet events, which turned out to be much higher than the static-domain rate; we kept it rather than re-select after seeing the twin results. Scenes are not independent draws: several LASIESTA sequences share a room or setup, and the site-level bound of Section 6.4 exceeds $\varepsilon/2$. The cost comparison counts real events and CPU-hours only; annotating the calibration events and building plates are additional costs. Finally, an event miss is a component-level failure; linking it to a system hazard requires a hazard analysis of the application.

Independence and phase. The results assume independent events given the scene and independent scenes. Events of one camera share a background, a detector and, in deployment, a single sampling phase; benchmark scenes are not a random sample of deployments. Correlation of events within a scene does not invalidate the population bound of Result 4(a) or the leave-one-scene-out audit, because both count scenes, not events: a scene contributes one indicator whatever the dependence among its events. It does affect the fleet bound of Result 4(b), which counts events and becomes optimistic under positive within-scene dependence; the population bound needs only independence across scenes. The main results certify the phase-averaged miss rate;

the worst-phase convention of Section 6.4 covers any fixed phase at a higher real-event cost.

Threats to validity. The audit tables and figures use level δ, while acceptance uses $\delta/2$ for the transfer term; at $\delta/2$ the bounds are larger (Result 4(a)). Software: Python 3.11.9, NumPy 1.26.4, SciPy 1.16.3, pandas 2.3.3, Ultralytics 8.4.7, OpenCV 4.10.0; CPU: Intel Core i7-1185G7. The detector, the event definition and the hit rule are fixed; a different detector changes p, q and D but not the theorems. The number of scenes, not the number of events, limits the transfer certificate: the static domain reaches $U(J, m) = 6.05\%$ with no discordant scene among 48, and more scenes are needed for tighter bounds. The audit at OP* is weak because real misses are rare there. The label-based static domain was fixed after the initial version of this study (Section 5); its clean audit is therefore partly a result of that post-hoc choice.

Practical guidance. The results suggest a simple decision rule for practitioners who consider replacing a field acceptance test with a twin. (i) Compute n_0 at the target (ε, δ) and estimate the real miss rate; if it is far below ε, the direct test is already short and a twin can save at most n_0 events. (ii) If a twin is built, declare its domain before calibration and calibrate discordance on as many scenes of that domain as possible; the number of scenes sets the resolution (Result 4(a)). (iii) Run each new camera's twin long enough to bound its own miss rate; three runs per event, as here, are not enough. (iv) Do not use the transfer certificate to admit a camera whose conditions differ from the calibration domain (night, a shaking mount): the certificate is withdrawn when such scenes enter the calibration, but a single such camera passes through.

8. Conclusion

Without an assumption linking simulation to reality, synthetic evidence buys no real events in zero-failure acceptance testing, and relaxing the error level buys an exactly computable number. A paired twin turns the question into one of discordance. The population bound is the transfer guarantee: it counts scenes, tolerates dependence within a scene, and its price is the number of scenes; the fleet bound covers only the calibrated cameras and assumes independent events. On the 48 static scenes of this study the transfer certificate stands at scene level ($U(J, m) = 6.05\%$) but is withdrawn at site level, where 17 sites give 16.2%, above $\varepsilon/2$. Measured on public benchmarks,

the exchange rate is modest and domain-bound; at $\varepsilon = 5\%$ the transfer needs at least 119 calibration scenes. The certificate has two named failure modes, night and camera jitter (the latter seen in one scene with real shake), in which out-of-domain cameras pass through unless the domain is declared. In the terms of simulation validation, the discordance D is a decision-specific validation metric of the composite model, the number of discordance-free calibration scenes is the validation evidence the acceptance decision requires, and the domain rule is an applicability check [24]; the credibility of the model is then expressed in field events replaced rather than as a level or a score [2, 5].

CRediT authorship contribution statement

Lam Quoc Dat: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares no competing interests. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic), an AI assistant, in order to write analysis and plotting scripts, to check the numbers of the manuscript against the result files, and to draft and edit text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data availability

Code, event manifests, result tables and the CC0 person-sprite set are publicly available at `https://github.com/lamquocdat01/no-free-credit-ress-artifact` (tag ress-v1.2, commit 890465bd9267). Raw video frames (CDnet 2014, LASIESTA, BMC) and Virtual KITTI 2 sprites are not redistributed under their licences; the repository provides download instructions and SHA-256 checksums to regenerate every derived file.

Appendix A. Formal statements and proofs

This appendix states the results of Section 4 formally and proves them. Theorem A.1 is Result 1, Theorem A.3 is Result 2, Lemma A.5 and Proposition A.15 are Result 3, and Proposition A.11, Theorem A.9 and Proposition A.13 are Result 4(a), (b) and (c).

Appendix A.1. Setting and notation

Events and misses. A camera (scene) s produces events $i = 1, 2, \dots$ (a maximal run of frames containing the target object). The system samples frames periodically with period K and an unknown phase $\varphi \in \{0, \dots, K-1\}$; a detector is run on the sampled frames. An event is *missed* ($M = 1$) when no sampled frame of the event is a hit (the detector finds the object). With $H \subseteq \mathbb{Z}$ the hit frames of the event and φ uniform, $\mathbb{P}(M = 1 \mid H) = 1 - |H \bmod K|/K$, where $H \bmod K$ is the set of residues of H. The miss rate of camera s is $p_s = \mathbb{P}(M = 1)$ over its events and phases. Target: certify $p_s \le \varepsilon$ with confidence $1 - \delta$.

Clopper–Pearson bound. For $x \in \{0, \dots, n\}$, $F_{n,p}(x) = \mathbb{P}(\mathrm{Bin}(n, p) \le x)$ and $U(x, n) = U_\delta(x, n)$ is the one-sided $(1-\delta)$ upper Clopper–Pearson bound [71]: the solution u of $F_{n,u}(x) = \delta$ for $x < n$, and 1 for $x = n$. Since $F_{n,p}(x)$ is strictly decreasing in p, $U(x, n) < p \iff F_{n,p}(x) < \delta$, and U is non-decreasing in x. Validity: $\mathbb{P}(U(\mathrm{Bin}(n, p), n) < p) \le \delta$ for every p.

Success run. $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1-\varepsilon) \rceil$ is the smallest n with $(1-\varepsilon)^n \le \delta$ (e.g. $n_0(5\%, 5\%) = 59$, $n_0(20\%, 5\%) = 14$).

Simulator and twin. Simulator data Y have a law Q on an arbitrary space. A *twin* is a paired simulator: every real event i has a synthetic counterpart with miss indicator M_i' under the same sampling phase. $q = \mathbb{P}(M' = 1)$, and the *discordance* is $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch).

Appendix A.2. C1: no free credit

A certification procedure is a (possibly randomised) test $\phi(X, Y) \in \{0, 1\}$ with real data $X = (M_1, \dots, M_n) \sim \mathrm{Bern}(p)^{\otimes n}$ independent of simulator data $Y \sim Q$. It is *valid at level δ on a class $\mathcal{P}$* of pairs (p, Q) if $\sup_{(p,Q) \in \mathcal{P},\, p > \varepsilon} \mathbb{P}_{p,Q}(\phi = 1) \le \delta$.

Theorem A.1 (C1)**.** *Assume $\mathcal{P}$ places no restriction linking Q and p: for every Q there are $(0, Q) \in \mathcal{P}$ and $(p, Q) \in \mathcal{P}$ for all $p \in (\varepsilon, \varepsilon + \eta)$ for some*

$\eta > 0$. *If ϕ is valid at level δ on $\mathcal{P}$, then for every Q and every simulator sample size,*

$$(1-\varepsilon)^n \, \mathbb{P}_{0,Q}(\phi = 1) \le \delta.$$

In particular, certifying a perfect system ($p = 0$) with probability one requires $n \ge n_0(\varepsilon, \delta)$ real events, whatever the amount of simulator data; power $1-\beta$ at $p = 0$ requires $n \ge \ln(\delta/(1-\beta))/\ln(1-\varepsilon)$.

Proof. Fix Q and write $g(y) = \mathbb{E}[\phi(0^n, y)]$ (expectation over the internal randomisation). Under $p = 0$, $X = 0^n$ almost surely, so $\mathbb{P}_{0,Q}(\phi = 1) = \mathbb{E}_Q[g(Y)]$. Under $p > \varepsilon$, the event $\{X = 0^n\}$ has probability $(1-p)^n$ and is independent of Y, hence $\mathbb{P}_{p,Q}(\phi = 1) \ge (1-p)^n \, \mathbb{E}_Q[g(Y)] = (1-p)^n \mathbb{P}_{0,Q}(\phi = 1)$. Validity gives $(1-p)^n \mathbb{P}_{0,Q}(\phi = 1) \le \delta$ for all $p \in (\varepsilon, \varepsilon + \eta)$; letting $p \downarrow \varepsilon$ proves the inequality. With $\mathbb{P}_{0,Q}(\phi = 1) = 1$ it reads $(1-\varepsilon)^n \le \delta$, i.e. $n \ge n_0$; with $\mathbb{P}_{0,Q}(\phi = 1) \ge 1 - \beta$ it gives the power bound. □

Remark A.2. The argument only uses that the simulator data have the same law under the two hypotheses and that the all-success real sample has positive probability under both. It is the success-run analogue of the observation that external data cannot increase power under strict type-I error control over all external-data laws [22]. *Check.* The zero-failure test with $n = 59$ at $p = \varepsilon = 5\%$: $(0.95)^{59} = 0.0485$ (MC 0.0491, 10^5 runs).

Appendix A.3. C2: the price of borrowing

Suppose one accepts a two-tier guarantee: level δ when the simulator is "right" and only level $\delta' > \delta$ in general.

Theorem A.3 (C2, exact credit)**.** *(i)* Converse. *Under the assumption of Theorem A.1, any procedure valid at level δ' on $\mathcal{P}$ that certifies a perfect system with probability one uses $n \ge n_0(\varepsilon, \delta')$ real events. Hence the credit, the number of real events saved with respect to $n_0(\varepsilon, \delta)$, is at most*

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta').$$

(ii) Attainment. *Let $T(Y) \in \{0, 1\}$ be any simulator test ("the simulator passes") and let B be the procedure: if $T = 1$ run the zero-failure test with $n_1 = n_0(\varepsilon, \delta')$ real events, otherwise with $n_0 = n_0(\varepsilon, \delta)$. Then B is valid at level δ' on $\mathcal{P}$, saves exactly c events whenever $T = 1$, and is valid at level δ on the subclass*

$$\mathcal{P}_0 = \Big\{(p, Q) : \; p > \varepsilon \Rightarrow \mathbb{P}_Q(T = 1) \le \pi^*\Big\}, \qquad \pi^* = \frac{\delta - (1-\varepsilon)^{n_0}}{(1-\varepsilon)^{n_1} - (1-\varepsilon)^{n_0}},$$

i.e. where a simulator of a system that is truly worse than ε rarely passes. *(iii)* Continuous lower bound. $c \geq c^* = \lfloor \ln(\delta'/\delta)/(-\ln(1-\varepsilon)) \rfloor$.

Proof. (i) Theorem A.1 at level δ' gives $(1-\varepsilon)^n \leq \delta'$, whose smallest integer solution is $n_0(\varepsilon, \delta')$. (ii) For $p > \varepsilon$, with $\pi = \mathbb{P}_Q(T = 1)$ and T independent of the real data, $\mathbb{P}(B = 1) = \pi(1-p)^{n_1} + (1-\pi)(1-p)^{n_0} \leq (1-\varepsilon)^{n_1} \leq \delta'$, and $\leq \delta$ as soon as $\pi \leq \pi^*$ (the expression is affine in π and decreasing in p). (iii) Write $a = \ln \delta / \ln(1-\varepsilon)$, $a' = \ln \delta' / \ln(1-\varepsilon)$ and $r = a - a' = \ln(\delta'/\delta)/(-\ln(1-\varepsilon))$. Since $n_0(\delta) \geq a$ and $n_0(\delta') < a' + 1$, $c > r - 1$; c being an integer, $c \geq \lfloor r \rfloor = c^*$. □

Remark A.4 (numbers and checks). At $\varepsilon = \delta = 5\%$: $\delta' = 7.5, 10, 15, 20\%$ give $c = 8, 14, 22, 27$ ($-14\%, -24\%, -37\%, -46\%$ of 59). The exact form is maximal on all 40 grid points ($(1-\varepsilon)^{n_0(\delta')} \leq \delta' < (1-\varepsilon)^{n_0(\delta')-1}$: yes); MC ($10^5$ runs per point) matches the closed form within 0.29 percentage points. *Empirical C2* (procedure B, $T =$ the one-sided CP bound of the simulator's miss rate $\leq \varepsilon$, 10^5 runs, $\varepsilon = 5\%$, $d_{\min} = 32$, $K = 1$): with the paired twin of the 48 static scenes (114 events) the simulator passes in 73.3% of runs; realised credit 5.9/10.3/16.1/19.8 events and false certification at the worst case $p = \varepsilon$ equal to 6.7%/8.4%/12.2%/15.6% for $\delta' = 7.5/10/15/20\%$ — the credit is bought exactly with the extra error $\delta' - \delta$. The off-the-shelf synthetic benchmark (30 events of length ≥ 32) never passes (0.0%): too few synthetic events to bound its own miss rate below 5%.

Appendix A.4. C3: the twin lemma

Lemma A.5 (C3)**.** *For any pairing (M, M') of a real event and its twin, $p \leq q + D$ with $D = \mathbb{P}(M = 1, M' = 0)$. No assumption links the twin to reality.*

Proof. $p = \mathbb{P}(M = 1, M' = 1) + \mathbb{P}(M = 1, M' = 0) \leq \mathbb{P}(M' = 1) + D = q + D$. □

Corollary A.6 (a per-camera twin does not save events)**.** *If $q \leq q_U$ holds with confidence $1 - \delta/2$ and $D \leq \varepsilon - q_U$ is certified for the same camera by a zero-discordance success run at level $\delta/2$, the number of paired real events is $\lceil \ln(\delta/2)/\ln(1-(\varepsilon - q_U)) \rceil = 72, 91, 122$ for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$, all larger than $n_0 = 59$. The twin can only help when D is calibrated once, on many scenes, and transferred (C4).*

Remark A.7 (event unit). With a uniform phase, $x_i = \mathbb{P}_\varphi(M_i = 1, M_i' = 0) = |R_i' \setminus R_i|/K$, where R_i, R_i' are the residue sets of the real and twin hit frames of event i (twin catch decided by the majority of three sprite variants). The *worst-phase* indicator $W_i = \mathbf{1}\{R_i' \setminus R_i \neq \emptyset\}$ satisfies $\mathbb{P}(M_i = 1, M_i' = 0 \mid \text{event}) = x_i \leq W_i$. All certificates below are stated for Bernoulli indicators Z_i; they hold verbatim with $Z_i = W_i$ and then bound the (larger) worst-phase discordance rate. This integer form is the one used for the reported certificates.

Appendix A.5. fleet bound: event-weighted certificate for the calibrated fleet

Lemma A.8 (mean–median)**.** *For $N \geq 2$ and $0 < p < 1 - 1/N$, $F_{N,p}(\lfloor Np \rfloor) > 1/4$. Consequently, if $\delta \leq 1/4$ and $F_{N,p}(x) < \delta$, then $x \leq Np - 1$.*

Proof. Greenberg and Mohri [63] prove $\mathbb{P}(B \geq \mathbb{E}B) > 1/4$ for $B \sim \mathrm{Bin}(N, r)$ with $r > 1/N$. Apply it to $B = N - \mathrm{Bin}(N, p) \sim \mathrm{Bin}(N, 1-p)$ with $1 - p > 1/N$: $\mathbb{P}(\mathrm{Bin}(N, p) \leq Np) > 1/4$, and $\{\mathrm{Bin}(N, p) \leq Np\} = \{\mathrm{Bin}(N, p) \leq \lfloor Np \rfloor\}$. For the consequence: $F_{N,p}$ is non-decreasing, so $F_{N,p}(x) < \delta \leq 1/4 < F_{N,p}(\lfloor Np \rfloor)$ forces $x < \lfloor Np \rfloor$, i.e. $x \leq \lfloor Np \rfloor - 1 \leq Np - 1$. *Check:* $\inf F_{N,p}(\lfloor Np \rfloor)$ over $N = 2..2000$ and a fine grid of p is 0.2500, approached at $N = 2$, $p \to 1/2$ (the constant $1/4$ is tight). □

Theorem A.9 (fleet bound)**.** *Let $Z_1, \ldots, Z_N$ be independent, $Z_i \sim \mathrm{Bern}(d_i)$ with arbitrary $d_i \in [0, 1]$ (heterogeneous across scenes and events), $X = \sum_i Z_i$ and $\bar{d} = \frac{1}{N}\sum_i d_i$. If $\delta \leq 1/4$ and $\bar{d} < 1 - 1/N$, then*

$$\mathbb{P}\big(U(X, N) < \bar{d}\big) \leq \delta.$$

With $d_i = D_{s(i)}$ (the discordance of the scene of event i), $\bar{d} = \bar{D}_w = \sum_s n_s D_s / N$, the event-weighted discordance of the calibrated cameras; combined with Lemma A.5, $\bar{p}_w \leq \bar{q}_w + U(X, N)$ with probability at least $1 - \delta$.

Proof. If $\bar{d} = 0$ there is nothing to prove. Otherwise let $x^* = \max\{x : F_{N,\bar{d}}(x) < \delta\}$ (if the set is empty the failure probability is 0). By monotonicity of U, $\{U(X, N) < \bar{d}\} = \{X \leq x^*\}$. By Lemma A.8, $x^* \leq N\bar{d} - 1$. Hoeffding's Theorem 4 [62]: for a sum S of independent Bernoulli variables with mean $N\bar{d}$ and $0 \leq b \leq N\bar{d} - 1$, $\mathbb{P}(S \leq b) \leq F_{N,\bar{d}}(b)$. With $b = x^*$, $\mathbb{P}(X \leq x^*) \leq F_{N,\bar{d}}(x^*) < \delta$. □

Remark A.10 (what it does not say; checks). The guarantee is conditional on the calibrated scenes and weighted by events: it does not bound any single camera and does not transfer to a new camera (if scenes are random, X is over-dispersed; e.g. $D_s \in \{0,1\}$ gives $\mathbb{P}(X=0) = (1-\mu)^m \gg (1-\mu)^N$). *Checks.* Six synthetic heterogeneity patterns on a fixed configuration of 44 scenes with $N = 304$ events (a calibration set from an earlier stage of the study, used only as a test bed for the theorem): exact Poisson–binomial failure ≤ 0.0471, MC (10^5) ≤ 0.0473. Measured discordance probabilities of the study (event level, main domain, five operating points, as measured and rescaled to mean 2% and 5%; 12 cases): exact failure ≤ 0.0272, MC ≤ 0.0267, all $\leq \delta = 0.05$.

Appendix A.6. population bound: transfer to a new camera

Proposition A.11 (population bound)**.** *Let scenes $S_1, \ldots, S_m$ be i.i.d. from a scene population, each with $n_s \geq 1$ paired events whose discordance indicators are independent* $\mathrm{Bern}(D_{S_s})$ *given the scene, $\mu = \mathbb{E}[D_S]$, and $J = \#\{s : X_s \geq 1\}$ with X_s the number of discordant events in scene s. Then*

$$\mathbb{P}\big(U(J,m) < \mu\big) \leq \delta.$$

For a new camera drawn from the same population, $\mathbb{E}[p_{\mathrm{new}}] \leq \mathbb{E}[q_{\mathrm{new}}] + U(J,m)$ with probability $\geq 1-\delta$ over the calibration, and $\mathbb{P}(D_{\mathrm{new}} > t) \leq U(J,m)/t$ (Markov).

Proof. Let $B_s = \mathbf{1}\{X_s \geq 1\}$. Given the scene, $\mathbb{P}(B_s = 1 \mid S_s) = 1 - (1 - D_{S_s})^{n_s} \geq D_{S_s}$ because $(1-D)^n \leq 1 - D$ for $n \geq 1$. Hence $\beta_s := \mathbb{P}(B_s = 1) \geq \mu$, and the B_s are independent across scenes. By a coupling $(B_s = \mathbf{1}\{V_s \leq \beta_s\} \geq \mathbf{1}\{V_s \leq \mu\}$ with i.i.d. uniforms V_s), J stochastically dominates $\mathrm{Bin}(m, \mu)$. Since $U(\cdot, m)$ is non-decreasing, $\mathbb{P}(U(J,m) < \mu) \leq \mathbb{P}(U(\mathrm{Bin}(m,\mu), m) < \mu) \leq \delta$ by Clopper–Pearson validity. The corollary follows from Lemma A.5 applied to the new camera and averaged over the population. □

Remark A.12 (price and checks). The resolution is set by the number of scenes: with $J = 0$, $U \leq 10\%$ needs $m \geq 29$ and $U \leq 5\%$ needs $m \geq 59$. *Checks.* The tight case $D_S \in \{0,1\}$, $n_s = 1$, $m = 48$ (then $J \sim \mathrm{Bin}(m,\mu)$ exactly), $\mu \in \{1, 2, 5, 10, 20\}\%$, 10^5 runs: maximum failure 0.0413. Leave-one-scene-out on the 48 static scenes at the primary operating point ($d_{\min} = 32$, $K = 16$): $D_s > U(J_{-s}, m-1)$ for 0 of 48 scenes (the proposition bounds the population mean, not each scene).

Appendix A.7. per-camera form: a per-camera (quantile) certificate and its data requirement

Proposition A.13 (per-camera form)**.** *Under the assumptions of Proposition A.11 with $n_s \geq n_{\min}$ for all s, let $t \in (0,1)$, $\pi_t = \mathbb{P}(D_S > t)$ and $h_t = 1-(1-t)^{n_{\min}}$. Then*

$$\mathbb{P}\big(\pi_t > U(J,m)/h_t\big) \leq \delta.$$

Hence, with $\hat{\gamma} = U(J,m)/h_t$, a new camera satisfies $p_{\text{new}} \leq q_{\text{new}} + t$ except on a set of cameras of population mass at most $\hat{\gamma}$, with confidence $1-\delta$.

Proof. $\mathbb{P}(B_s = 1) \geq \mathbb{P}(D_S > t)\big(1-(1-t)^{n_s}\big) \geq \pi_t h_t$. The coupling argument of Proposition A.11 gives $J \succeq_{\text{st}} \text{Bin}(m, \pi_t h_t)$, so $\mathbb{P}(U(J,m) < \pi_t h_t) \leq \delta$, which is the claim. □

Remark A.14 (data requirement; checks). With $J = 0$, $\hat{\gamma} \leq \gamma$ requires $m \geq m_{\min}$ with $1-\delta^{1/m_{\min}} \leq \gamma$ ($m_{\min} = 59$ for $\gamma = 5\%$, 29 for $\gamma = 10\%$) and $n_{\min} \geq \ln\big(1-U(0,m)/\gamma\big)/\ln(1-t)$ events in every scene; the total $m\, n_{\min}$ decreases to the floor $\approx \ln(1/\delta)/(\gamma t)$ as m grows. *Check:* two-point populations $D_S \in \{0, t'\}$, $t' > t$, $m = 48$, 10^5 runs: maximum failure 0.0442.

Appendix A.8. C5a: a pessimistic-length twin is conservative under periodic sampling

Proposition A.15 (C5a)**.** *Let an event occupy frames $F = \{o, \ldots, o+d-1\}$ and let the detector outcome on each frame be fixed (a deterministic function of the frame). For the truncated event $F_L = \{o, \ldots, o+L-1\}$, $L \leq d$, with hit set $H_L = H \cap F_L$, the miss probability under a uniform phase, $m(L) = 1-|H_L \bmod K|/K$, is non-increasing in L. Consequently a twin built with a pessimistic (shorter) length L has $q_L \geq q$ and discordance $D_L \leq D$, so the C3 bound $q_L + D_L$ remains valid and the miss term is conservative.*

Proof. $L \leq L'$ implies $H_L \subseteq H_{L'}$, hence $H_L \bmod K \subseteq H_{L'} \bmod K$ and $m(L) \geq m(L')$. A twin catch at phase φ for the truncated twin implies a catch for the full twin, so $\{M = 1, M'_L = 0\} \subseteq \{M = 1, M' = 0\}$ and $D_L \leq D$; Lemma A.5 holds for any pairing. □

Remark A.16 (check). On the twin of the 48 static scenes, shortening every event by $x \in \{10, 25, 50\}\%$ gives non-decreasing q for every event and every $K \in \{1, 4, 8, 16, 48\}$ (all monotone: yes); e.g. at $d_{\min} = 32$, $K = 16$, q rises from 1.37% to 2.36% at $x = 50\%$.

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Table 1: Research gap. Credibility: how the worth of simulation evidence is expressed. ZF: is the zero-failure acceptance test treated? Field events: is the number of field observations that simulation replaces quantified? Transfer: does the guarantee reach an untested unit? Link: assumption needed between simulation and reality. Evidence: empirical basis.

Body of work	Credibility expressed as	ZF	Field events replaced	Transfer	Sim–real link	Evidence
(a) Simulation validation and credibility [2, 3, 5, 8, 7, 24, 9]	level, score or predictive uncertainty	no	not quantified	applicability to a new context of use argued, not bounded	validation against field data; discrepancy model	case studies
(b) Simulation-assisted testing of automated and perception systems [15, 16, 25, 19, 26, 27]	gap measured; variance reduction for means	no	fewer real samples for a mean metric	not addressed (extrapolation named as open)	paired outputs (control variate, rectifier) or none	benchmarks, case studies
(c) Zero-failure test planning [10, 28, 29, 12, 13, 30]	prior or auxiliary information	yes	through a prior; not as field events saved	to the tested unit; prior-system evidence handled conservatively	prior or similarity assumption (Bayesian); none (frequentist)	case studies
(d) Finite-sample guarantees and assurance for learned perception [31, 32, 33, 34, 35]	not addressed (real calibration data)	no	none	exchangeability of calibration and deployment	exchangeability	detection data sets
(e) This paper	field events saved per unit	yes	exact: none without a link; $n_0(\delta) - n_0(\delta')$ when relaxing; 14 events per camera at most	yes at scene level (withdrawn at site level), priced by scene count ($U(J, m) = 6.05\%$, $J = 0$, $m = 48$); scenes exchangeable (i.i.d.), as in (d) at scene level	paired twin + measured discordance D	48 fixed-camera scenes, 114 events; failure mode measured

Table 2: Notation.

symbol	meaning
$\varepsilon,\ \delta$	target miss probability; allowed probability of a false acceptance
$n_0(\varepsilon, \delta)$	zero-failure sample size $\lceil \ln \delta / \ln(1-\varepsilon) \rceil$
$K,\ \varphi,\ d_{\min}$	sampling period, sampling phase, minimum event length (frames)
$M,\ M'$	real miss, twin miss of an event at the same phase
$p_s,\ q_s,\ D_s$	real miss probability, twin miss probability, discordance $\mathbb{P}(M=1, M'=0)$ of camera s
$m,\ n_s,\ N$	calibration scenes, paired events of scene s, $N = \sum_s n_s$
$X,\ J$	discordant events; scenes with at least one discordant event
$U(x, n)$	one-sided $(1-\delta)$ Clopper–Pearson upper bound
fleet bound, population bound	$U(X, N)$ and $U(J, m)$ (Result 4)
OP*, OP-A, OP-C	operating points $(\varepsilon, d_{\min}, K) = (20\%, 32, 16)$, $(20\%, 32, 8)$, $(10\%, 32, 1)$

Table 3: Result 2: exact credit c versus the continuous form c^* ($\varepsilon = \delta = 5\%$, $n_0 = 59$), and the two-tier plan run with the twin event pool as an independent simulator (worst case $p = \varepsilon$; closed form given the pass probability, and 100,000 Monte-Carlo runs).

δ'	7.5%	10%	15%	20%
exact credit c (demands)	8	14	22	27
saving w.r.t. n_0	14%	24%	37%	46%
continuous form c^*	7	13	21	27
false acceptance, twin pool (closed form)	6.7%	8.6%	12.3%	15.5%
Monte Carlo	6.7%	8.4%	12.2%	15.6%

Table 4: Data needed for the per-camera form of Result 4(c) with no discordance ($J = 0$): minimum number of scenes $m_{\min}$, events per scene n at $m = m_{\min}$ and at $m = 2m_{\min}$, total events, and the asymptotic floor $\ln(1/\delta)/(\gamma t)$.

t	γ	$m_{\min}$	n at $m_{\min}$	n at $2m_{\min}$	total at $2m_{\min}$	floor
2.5%	5%	59	183	28, 3,304	2,397	
2.5%	10%	29	158	28, 1,624	1,198	
5%	5%	59	91	14, 1,652	1,198	
5%	10%	29	78	14, 812	599	
10%	5%	59	44	7, 826	599	
10%	10%	29	38	7, 406	300	

Table 5: Certificates by tier at OP* ($\delta = 5\%$). Discordant events are counted at the worst sampling phase (in parentheses: phase-averaged count, mean over the three sprite variants; per-scene tables use the majority rule, which gives 8.31 at night); $\hat{p}$, $\hat{q}$: real and twin miss rates averaged over the phases / at the worst phase (any fixed deployment phase); J = scenes with a discordant event.

tier	events	$\hat{p}$ avg/worst	$\hat{q}$ avg/worst	discordant	$U(X, N)$	J/m	$U(J, m)$
static (main)	114	0.9%/0.9%	1.4%/3.5%	0 (0.0000)	2.59%	0/48	6.05%
camera jitter	62	0.5%/4.8%	0.0%/0.0%	3	12.0%	1/26	17.0%
night	25	36.2%/52.0%	4.8%/16.0%	12 (8.21)	65.9%	5/6	99.1%
turbulence	2	6.2%/50.0%	0.0%/0.0%	1	97.5%	1/2	97.5%
all tiers	203	5.2%/8.9%	1.4%/3.9%	16	11.7%	7/82	15.4%

Table 6: Exchange rate at three operating points (48 static cameras). n_{dir}: real events of the smallest direct plan (failures allowed, Clopper–Pearson acceptance) with 80% power at the pooled static-domain $\hat{p}$; the zero-failure plan alone needs n_0 events; saved: median real events saved per camera by a twin run to convergence; runs: twin runs needed per camera; CPU-h: CPU-hours per saved real event.

point	$\hat{p}$	n_{dir}	saved	cams	runs	CPU-h
OP* $(20\%, 32, 16)$	0.88%	14	14	46	28	0.069
OP-A $(20\%, 32, 8)$	0.88%	14	14	47	28	0.064
OP-C $(10\%, 32, 1)$	0.88%	46	46	47	149	0.104
worst-phase convention (valid for any fixed phase):						
OP*	0.9%	14	14	44	28	–
twin as built: cameras saving = 3, 3, 0 (median saving 0)						

Table 7: Operating-point grid at $\varepsilon = 20\%$ (condensed; full grid in the supplement). Each cell: static-domain real miss rate $\hat{p}$ / real events for direct certification / median real events saved per camera by a twin run to convergence. The median saving of the twin as built is 0 at every cell. OP* is $(32, 16)$.

$d_{\min} \backslash K$	1	8	16	24	48
32	0.9%/14/14	0.9%/14/14	0.9%/14/14	1.2%/14/14	5.9%/37/0
24	0.8%/14/14	1.1%/14/14	1.7%/22/22	2.3%/22/22	8.2%/57/0
16	1.6%/14/14	2.6%/22/22	4.3%/30/30	6.1%/37/37	13.5%/210/0
12	2.3%/22/22	3.4%/22/22	5.0%/30/30	6.8%/44/44	14.1%/260/0
8	2.2%/22/22	4.0%/30/30	6.9%/44/44	9.2%/69/69	16.8%/901/0
4	2.7%/22/22	8.4%/63/63	12.8%/170/170	15.4%/441/441	23.0%/–/0
1	3.1%/22/22	14.6%/310/310	19.5%/–/0	22.2%/–/0	29.4%/–/0

Table 8: Real versus twin outcomes (expected event counts under a uniform sampling phase; twin = majority of three variants). Recall: share of expected real misses that the twin also misses.

	static, OP*	night, OP*
real miss, twin miss	1.00	0.75
real miss, twin catch (D)	0.00	8.31
real catch, twin miss	0.56	0.44
real catch, twin catch	112.44	15.50
events	114	25
events missed at some phase	1	13
twin recall of real misses	100%	8%

Table 9: Must-fail scenarios. Top: temporal ($d_{\min} = 1$, $K = 48$), static scenes with $p_s > \varepsilon$, real misses split into detector and temporal misses; all refused in every Monte-Carlo run. Middle: night scenes at OP* (every miss is a detector miss), and whether each is falsely certified when judged alone (phase-averaged); then the same for the camera-jitter scenes (CDnet scenes with real shake listed; the LASIESTA scenes with simulated camera motion pooled, event-weighted). Bottom: population bound when k night scenes leak into the calibration set (median over subsets; withdrawn = share of subsets with a bound above $\varepsilon/2$).

1: temporal scene	n	p_s	detector	temporal	q_s	refused
port_0_17fps	5	99.6%	43.3%	56.2%	100.0%	100%
tramCrossroad_1fps	11	77.7%	6.1%	71.6%	74.2%	100%
tunnelExit_0_35fps	34	83.6%	5.2%	78.4%	87.9%	100%

2: detector (night) scene	n	p_s	q_s	D_s	disc.	false cert.
bridgeEntry	2	15.6%	0.0%	15.6%	1	no
busyBoulvard	4	96.9%	0.0%	96.9%	4	yes
fluidHighway	4	26.6%	0.0%	26.6%	2	yes
streetCornerAtNight	10	28.1%	11.9%	20.6%	4	yes
tramStation	2	0.0%	0.0%	0.0%	0	no
winterStreet	3	33.3%	0.0%	33.3%	1	yes
3: detector (camera jitter)						
boulevard	8	0.0%	0.0%	0.0%	0	no
traffic	8	3.9%	0.0%	3.9%	3	no
LASIESTA SM (24 scenes)	46	0.0%	0.0%	0.0%	0	0 of 24

k leaked	0	1	2	3	4	6
U_{pop}	6.1%	9.3%	12.1%	13.2%	14.2%	18.5%
withdrawn	0%	0%	67%	100%	100%	100%

Table 10: Twin fidelity on the same frames (static domain): hit rates by object kind and sprite aspect error before the width clamp, and median detector confidence on hits. Negative differences make the twin pessimistic.

kind	|aspect error|	frames	twin hit	real hit	diff. (pts)
person	≤10%	8,870	91.7%	94.8%	-3.2
person	10-20%	5,899	90.1%	94.2%	-4.2
person	20-40%	4,404	74.0%	86.2%	-12.2
person	>40%	5,449	34.4%	44.5%	-10.0
vehicle	≤10%	4,105	86.9%	89.0%	-2.1
vehicle	10-20%	2,569	88.0%	90.2%	-2.2
vehicle	20-40%	3,087	87.6%	85.1%	+2.5
vehicle	>40%	3,881	64.9%	74.2%	-9.3
person	confidence		0.82	0.90	
vehicle	confidence		0.72	0.78	

Table 11: Event-definition sensitivity (static domain). Primary: largest object ≥ 256 px; original: every ground-truth event. Discordant events at the worst phase; LOSO: scenes above the bound computed without them.

definition	point	N	$\hat{p}$	disc.	$U(X, N)$	J / $U(J, m)$	LOSO
primary	OP*	114	0.88%	0	2.59%	0 / 6.1%	0
	OP-A	114	0.88%	0	2.59%	0 / 6.1%	0
	OP-C	114	0.88%	0	2.59%	0 / 6.1%	0
original	OP*	116	0.86%	0	2.55%	0 / 6.1%	0
	OP-A	116	0.86%	0	2.55%	0 / 6.1%	0
	OP-C	116	0.86%	0	2.55%	0 / 6.1%	0